

Homework # 10

1. (50 points) The activation energy of an isomerization reaction which obeys 1st-order kinetics is known to be 100 kJ/mol and the pre-exponential of reaction is 1×10^5 cm/min. When the isomerization reaction is run under strong pore diffusion limitations at 400 K, 500 g of catalyst are required to achieve 50% conversion.

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a) Write the mole balance for the packed-bed reactor.

b) Calculate how much catalyst is required to get 50% conversion if the temperature is raised to 425 K.

You should assume that there are no external mass transfer limitations, but that strong pore diffusion limitations are present at 425 K. The catalyst density (ρ_c) is 2.5 g/cm³ and the bed void fraction (ϵ) is 0.4.

$$\rightarrow \frac{dF_A}{dV} = -\eta K'' S_0 \rho_c (1-\epsilon) C_A$$

$$F_A = C_A v$$

$$\rightarrow \frac{dC_A}{dV} = - \frac{\eta K'' A \cdot S_0 \rho_c (1-\epsilon) C_A}{v} = -A V$$

$$\rightarrow \ln\left(\frac{C_A}{C_{A0}}\right) = - \frac{\eta K'' S_0 \rho_c (1-\epsilon) V}{v} = -A V$$

$$\rightarrow \ln\left(\frac{C_A}{C_{A0}}\right) = -A W (1-\epsilon) \rho_c \rightarrow \ln\left(\frac{50,000}{1 \times 10^5}\right) = -A (500)(2.5)(1-.40)$$

$$A = 0.000924$$

$$C_A = C_{A0}(1-x) = 1 \times 10^5 (1-.5) = 50,000$$

→ For 50% at 425 K, we need to find a new A
assume the only thing changing is the rate of reaction and η

→ calculate new K'' @ $T = 425$ K

$$A = \frac{A_2}{A_1} = \frac{K_2'' \eta_2}{K_1'' \eta_1} = \frac{K_2'' (K_1'')^{1/2}}{K_1'' (K_2'')^{1/2}}$$

$$K_2 = A e^{-E_a/RT_2} \quad K_1 = A e^{-E_a/RT_1}$$

$$\begin{aligned} \frac{K_1}{K_2} &= e^{-E_a/RT_1 + E_a/RT_2} \\ &= e^{-100 \times 10^3 / 8.314(400) + 100 \times 10^3 / 8.314(425)} \\ &= 0.17 \end{aligned}$$

$$\frac{W_2}{W_1} = \frac{K_1}{K_2} \Rightarrow W_2 = 0.17(500 \text{ g}) = 425 \text{ g}$$

2. (50 points) The irreversible gas-phase reaction



is carried out over a packed bed of solid catalyst particles. The reaction is first order in the concentration of A on the catalyst surface:

$$-r_A' = k' C_A$$

The feed consists of 50% (mole) A and 50% inerts and enters the bed at a temperature of 300 K.

The entering volumetric flow rate is 10 dm³/s (i.e. 10,000 cm³/s). The relationship between the

Sherwood number and the Reynolds number is:

$$Sh = 100 Re^{1/2}$$

As a first approximation, one may neglect pressure drop. The entering concentration of A is 1.0 M. Calculate the catalyst weight necessary to achieve 60% conversion of A for isothermal operation.

Additional Information:

Kinematic viscosity: $\nu = 0.02 \text{ cm}^2/\text{s}$

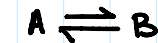
Particle diameter: $d_p = 0.15 \text{ cm}$

Superficial velocity: $U = 9 \text{ cm/s}$

Catalyst surface area/mass of catalyst bed: $a = 60 \text{ cm}^2/\text{g}$

Diffusivity of A: $D_c = 10^{-2} \text{ cm}^2/\text{s}$

$k' (300 \text{ K}) = 0.01 \text{ cm}^3/\text{s/g}$



$$-r_A' = k' C_A$$

$$Re = \frac{d_p U}{\nu}$$

$$Sh = 100 \cdot Re^{1/2}$$

$$X = .60$$

$$\text{if } Re = 9 \frac{\text{cm}}{\text{s}} \times \frac{0.15 \text{ cm}}{0.02 \text{ cm}^2/\text{s}} = 67.5$$

$$Sh = 100 (67.5)^{1/2} = 821.50$$

$$Sh = \frac{k_c \cdot d_p}{D_{AB}} \quad k_c = \frac{(10^{-2})(821.5)}{.15}$$

$$k_c = 54.76 \frac{\text{cm}}{\text{s}}$$

$$K_{eff} = \frac{k_c \cdot k''}{(k_c + k'')} \Rightarrow k'' = \frac{k'}{a} = \frac{.01 \text{ cm}^3/(\text{s} \cdot \text{g})}{60 \frac{\text{cm}^2}{\text{g}} \cdot \frac{1}{9}} = 1.67 \times 10^{-4} \frac{\text{cm}}{\text{s}}$$

$$K_{eff} = \frac{(54.76 \frac{\text{cm}}{\text{s}})(1.67 \times 10^{-4})}{54.76 + 1.67 \times 10^{-4}} = 1.67 \times 10^{-4}$$

$$\frac{dF_A}{dW} = K_{eff} \cdot C_A =$$

$$\ln\left(\frac{C_A}{C_{A0}}\right) = -\frac{W}{F_{A0}}(K_{eff}) \cdot a \rightarrow X = 0.6 \quad 1 - \frac{C_A}{C_{A0}} = 0.6$$

$$\frac{\ln(0.4) \times 10,000 \frac{\text{cm}^3}{\text{s}}}{(1.67 \times 10^{-4} \frac{\text{cm}}{\text{s}})(60 \frac{\text{cm}^2}{\text{g}})} = W = 2962.8 \text{ kg}$$